SwarpPay

Technical Whitepaper

A Solana-First Custodial Trading Platform
with Integrated Token Launchpad and Identity Verification

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1. Executive Summary

SwarpPay is a Solana-first, all-in-one custodial trading platform designed to bring mainstream users into the Solana ecosystem through a simplified, secure, and compliant interface. The platform addresses fundamental problems in today's fragmented cryptocurrency landscape: bot-dominated token launches, unsafe trading environments, complex user journeys across multiple applications, and the absence of per-person fairness guarantees.

The platform is built around three integrated components that work together to create a cohesive ecosystem:

SwarpPay: Custodial Wallet and Curated Trading

A custodial wallet experience where users can buy and sell supported crypto assets with fiat currency through embedded third-party on/off-ramp providers, then trade curated Solana tokens inside the same application. The custodial model eliminates private key management complexity for users while maintaining custody-grade operational controls including two-factor authentication, device binding, withdrawal risk rules, and anomaly detection.

SwarpLaunch: Request-Only Curated Launchpad

A token launchpad with standardized bonding curve mechanics and per-user participation caps enforced through identity verification. Projects submit launch requests and undergo review before acceptance. During the bonding phase, tokens are non-transferable, and each verified user is capped at 1% of total supply, preventing Sybil capture and ensuring fairer distribution. Upon reaching the graduation threshold of 169.97 SOL, liquidity automatically migrates to Meteora DAMM v2 for permissionless trading.

SwarpID: Verification and Anti-Duplicate Layer

An identity verification system that enforces one active verified human credential per person for gated actions including launch participation and higher-risk operations. SwarpID combines third-party KYC verification with internal anti-duplicate checks including name and address normalization, device fingerprinting, and risk scoring to prevent multiple account creation.

SwarpPay monetizes through a transparent fee structure: 0.50% platform fee per swap on listed tokens, 2.00% fee during SwarpLaunch bonding phase (split 1.50% to Swarp and 0.50% to creator), and post-graduation LP fee sharing from Meteora pools. Beginning in 2027, SwarpID will replace third-party providers and subsequently be commercialized as a B2B service for partners requiring per-person uniqueness verification.

The platform is pursuing a licensing-first regulatory strategy, targeting Estonia MiCA CASP authorization for EU market access and Dubai VARA VASP license for UAE operations. During the authorization process, SwarpPay will operate through licensed third-party partners under a defined outsourcing and controls framework.

2. Introduction and Problem Statement

2.1 Market Context

Solana has emerged as a leading blockchain platform for decentralized finance and token trading, offering high throughput, low transaction costs, and a vibrant ecosystem of projects and users. However, the current user experience for mainstream participants remains fragmented and hostile to retail users.

The typical Solana user journey requires navigating multiple disconnected applications: self-custody wallets for key management, fiat on-ramp services for currency conversion, decentralized exchanges for trading, token analytics platforms for research, and social channels for discovery. This fragmentation increases cognitive load, introduces security risks at each integration point, and creates opportunities for user error.

2.2 Core Problems

Bots and Sybil Capture Dominate Early Markets

Per-wallet fairness mechanisms allow sophisticated actors to control hundreds or thousands of wallets, capturing disproportionate allocations in token launches and distorting price discovery. Retail users consistently lose to automated systems that can execute faster and operate at scale.

Unsafe and Opaque Token Launches

Most token launches lack transparency around mechanics, team allocations, and liquidity arrangements. Hidden privileged wallets, concealed insider allocations, and liquidity traps are common. Users have no reliable way to distinguish legitimate projects from schemes designed to extract value.

Fragmented User Journey

Users must jump between wallets, DEX aggregators, chart platforms, token scanners, and social channels to complete a single trading decision. Each transition introduces friction, potential for error, and security exposure. The complexity creates barriers to entry for mainstream users.

No Consistent Trust Layer

Users struggle to distinguish legitimate projects from scams, and platforms lack standardized ways to enforce per-person limits rather than per-wallet limits. Without identity verification, fairness guarantees are technically unenforceable.

Compliance Readiness Gap

Many existing platforms were not designed with regulatory compliance in mind. As regulatory frameworks like MiCA in Europe and VARA in Dubai mature, platforms lacking proper controls, audit trails, and compliance infrastructure face existential risk.

2.3 SwarpPay Solution

SwarpPay addresses these problems through vertical integration of wallet, trading, launch, and identity services into a single platform with compliance readiness built in from day one. By combining custodial simplicity with curated listings, standardized launch mechanics, and per-person fairness enforcement, SwarpPay creates a safer environment for mainstream users to participate in the Solana ecosystem.

3. Platform Architecture Overview

3.1 System Components

SwarpPay is architected as a modular system with clear separation between client applications, backend services, blockchain infrastructure, and external integrations. The architecture prioritizes security, scalability, and regulatory compliance while maintaining a seamless user experience.

Client Applications

- Web Application: Next.js 15 with React 19, TypeScript, and Tailwind CSS
- Mobile Application: Flutter with BLoC state management, supporting iOS, Android, and web
- Both clients connect to a unified backend API

Backend Services

- API Server: NestJS framework with TypeScript on Node.js
- Database: PostgreSQL with TypeORM for relational data
- Authentication: JWT tokens with bcrypt password hashing
- Blockchain Integration: Solana web3.js for transaction construction and signing

Blockchain Layer

- Network: Solana (mainnet-beta for production, devnet for testing)
- RPC: Configurable third-party RPC providers with failover
- Token Standard: SPL Token and Token-2022 for advanced features
- DEX Integration: Jupiter aggregator and Raydium for swaps

External Integrations

- Identity Verification: Third-party KYC provider for user verification (SwarpID native in 2027)
- Fiat Rails: MoonPay for fiat currency conversion (SwarpID native in 2027)
- Communications: Email and SMS services for notifications and OTP
- Price Data: Multiple oracle sources for accurate pricing

3.2 Data Flow

User requests flow from client applications through the API layer, which handles authentication, validation, and business logic. Blockchain operations are constructed server-side and signed using custodial keys. Transaction status is monitored through polling mechanisms with confirmation tracked to finalized commitment level for maximum security.

For trading operations, the backend queries DEX aggregators for optimal routes, constructs transactions with appropriate slippage protection, signs using the user's custodial keypair, and submits to the Solana network. Users see real-time status updates as transactions progress through confirmation stages.

3.3 Security Architecture

Security is implemented in layers throughout the stack. Network-level protection includes DDoS mitigation at the infrastructure layer. Application-level security includes input validation using class-validator, parameterized database queries preventing SQL injection, and rate limiting on sensitive endpoints. Authentication uses JWT tokens with configurable expiry and refresh token rotation.

Custodial operations implement additional controls including PIN verification for high-risk actions, step-up authentication for large transactions, new withdrawal address cooldowns, and velocity-based anomaly detection that can trigger automatic freezes pending manual review.

4. SwarpPay: Custodial Wallet and Trading

4.1 Custodial Model

SwarpPay implements a full-custodial model where the platform generates, stores, and manages wallet private keys on behalf of users. Users do not receive seed phrases or direct access to private keys. This design choice prioritizes user experience and recoverability over self-sovereignty, targeting mainstream users who may not be comfortable with private key management.

Key Generation

User wallets are generated using Keypair.generate() from the Solana web3.js library, producing cryptographically random keypairs. Each user receives one primary wallet enforced at the database level. Keys are stored encrypted in the database using AES-256-CBC encryption with random initialization vectors.

Transaction Signing

When users initiate transactions, the backend retrieves the encrypted private key, reconstructs the keypair, constructs the transaction with current blockhash and appropriate compute budget, signs the transaction server-side, and submits to the Solana network. This flow is invisible to users who experience a simple trading interface.

4.2 Account Security

Authentication

Users authenticate through multiple channels including email, phone (SMS OTP), and social login (Google, Apple). All passwords and PINs are hashed using bcrypt with 10 salt rounds before storage. JWT tokens are issued upon successful authentication with configurable expiry periods.

Wallet PIN

Users set a 6-digit wallet PIN during onboarding. The PIN is required for high-risk operations including withdrawals and security setting changes. Five consecutive incorrect PIN attempts trigger a 30-minute account lock. PIN resets require re-verification through the original authentication channel.

Step-Up Authentication

Certain operations trigger step-up authentication requiring additional verification. Withdrawals above $10,000 equivalent require email OTP confirmation. First withdrawal to any new address requires step-up regardless of amount. Security setting changes including password reset, 2FA modification, and recovery initiate temporary withdrawal restrictions.

4.3 Withdrawal Controls

SwarpPay implements tiered withdrawal controls based on verification status and risk signals:

Tier 0: Unverified Users

- Daily withdrawal cap: $2,000 equivalent
- Monthly withdrawal cap: $20,000 equivalent
- 48-hour cooldown before first withdrawal to any new address
- 48-hour withdrawal lock after password reset, 2FA changes, or new device login
- Step-up authentication required for all withdrawals to new addresses

Tier 1: Verified Users

- No hard daily/monthly caps under normal conditions
- 24-hour cooldown for new withdrawal addresses
- 24-hour security change lock
- Step-up authentication required for withdrawals above $10,000
- Manual review triggered for withdrawals above $50,000

Risk-Based Overrides

Regardless of tier, certain patterns trigger automatic review or freeze: three withdrawal attempts within 10 minutes, three new addresses added within 24 hours, deposit followed by full withdrawal within 30 minutes on new accounts, multiple failed login or 2FA attempts, and risk score threshold breaches.

4.4 Trading Features

Curated Token Listings

SwarpPay maintains a curated list of tradeable tokens. Tokens are listed only after risk screening, liquidity verification, and market quality assessment. Users trade from this curated set, reducing exposure to scam tokens and low-quality projects that proliferate on permissionless DEXs.

Swap Execution

Swaps are routed through DEX aggregators (primarily Jupiter) to obtain optimal pricing. The platform displays clear fee disclosure separating SwarpPay's 0.50% platform fee from Solana network fees and underlying DEX pool fees. Slippage protection is configurable with sensible defaults.

Fiat On/Off-Ramp

Users can purchase crypto using fiat currency through the embedded MoonPay widget. Supported fiat currencies and crypto assets depend on MoonPay's availability in each jurisdiction. Fiat purchases credit directly to the user's SwarpPay wallet upon completion. Beginning in 2027, SwarpID will provide native fiat on/off-ramp infrastructure.

5. SwarpLaunch: Token Launchpad

5.1 Overview

SwarpLaunch is a curated, request-only token launchpad with standardized mechanics designed to reduce Sybil capture and increase transparency. Unlike permissionless launch

platforms, SwarpLaunch requires projects to submit applications and undergo review before acceptance. All launches follow a uniform template with fixed parameters, published disclosures, and per-person participation caps.

5.2 Launch Lifecycle

Phase 1: Application and Review

Project creators submit launch requests including token name, ticker, description, social links, and team information. SwarpPay evaluates legitimacy, token design, wallet disclosures, and project readiness. Acceptance is not pay-to-launch; quality and compliance are primary criteria.

Phase 2: Bonding

Approved projects enter the bonding phase where verified users can purchase tokens along a deterministic bonding curve. During bonding, tokens are non-transferable, preventing OTC deals and early dumping. Each verified user is capped at 1% of total supply (2% for designated developer wallets), enforcing per-person fairness rather than per-wallet fairness.

Phase 3: Graduation

When the bonding curve reaches its target raise of 169.97 SOL net, the launch graduates. Transfer restrictions are lifted, and liquidity automatically migrates to Meteora DAMM v2 for permissionless trading. Post-graduation, SwarpPay adds no additional platform fee for trades in graduated tokens.

5.3 Bonding Curve Mechanics

Fixed Launch Parameters

Parameter	Value
Total Token Supply	1,000,000,000
Bonding Sale Supply	793,000,000 (79.3%)
LP Reserve	207,000,000 (20.7%)
Graduation Target	169.97 SOL (net after fees)
Per-User Cap	1% of total supply
Developer Cap	2% of total supply
Creator Allocation	2% (locked during bonding)

Price Function

Token price during bonding is computed by a deterministic monotonic function of tokens sold. The curve is parameterized to reach the graduation target of 169.97 SOL net when 793,000,000 tokens have been sold. A linear bonding curve implementation uses the formula:

Price per token: $p(x) = p_0 + kx$

Where x is tokens sold, p_0 is starting price, and k is the slope parameter.

Total raised: $R(x) = p_0x + (1/2)kx^2$

Parameters p_0 and k are calculated such that R(793,000,000) equals the gross raise target (169.97 / 0.98 ≈ 173.44 SOL to account for 2% bonding fee).

5.4 Fee Structure

Bonding Phase Fees

- Total fee: 2.00% per transaction (buy and sell)
- Platform share: 1.50% to SwarpPay
- Creator share: 0.50% to project creator
- Fees deducted from SOL contribution

Post-Graduation Fees

After graduation to Meteora, trading fees are determined by the LP pool configuration (targeted around 1.00%). LP fee revenue is collected by the Swarp-controlled locked LP position and distributed: 70% to the creator and 30% to SwarpPay via automated distribution. SwarpPay adds no additional platform fee for trades in graduated SwarpLaunch tokens.

5.5 Transfer Restrictions

During bonding, tokens are non-transferable to prevent OTC bypass and early distribution manipulation. This restriction is enforced using Token-2022 transfer hooks that reject transfer instructions until the launch status transitions to graduated/migrated.

At graduation, the transfer hook is revoked, restoring normal SPL token transferability. All users can then freely transfer, trade, or use their tokens on any compatible platform.

5.6 Meteora Integration

Graduation deploys liquidity to Meteora DAMM v2 (Dynamic AMM). At graduation:

- All SOL raised during bonding (net of fees) pairs with the LP reserve tokens
- A new Meteora pool is created with the token/SOL pair
- The LP position is controlled by SwarpPay for fee collection and distribution
- Trading becomes permissionless on Meteora and any other DEX listing the token

6. SwarpID: Identity Verification

6.1 Purpose

SwarpID provides verification and anti-duplicate functionality that enforces one active verified human credential per person for gated actions. The system enables per-person fairness guarantees that are impossible with wallet-only identification, and supports stronger compliance and risk controls for fiat features.

6.2 Verification Tiers

Tier 0: Unverified

Users who have registered but not completed identity verification. Tier 0 users can trade curated tokens with daily and monthly withdrawal limits. They cannot participate in SwarpLaunch bonding events or access features requiring verified status.

Tier 1: Verified

Users who have completed identity verification through the third-party KYC provider. Verified users can participate in SwarpLaunch with per-person caps, access higher withdrawal limits, and qualify for enhanced rewards programs. Verification includes document verification, liveness check, and anti-duplicate screening.

6.3 Anti-Duplicate Detection

SwarpID implements multiple layers of duplicate detection to prevent single individuals from maintaining multiple verified accounts:

KYC Provider Checks

The third-party KYC provider performs identity document verification and maintains its own duplicate detection across its customer base.

Name and Address Normalization

- Case folding and diacritic removal
- Whitespace collapse and standardization
- Common transliteration handling
- Country-specific address format normalization
- Standard abbreviation expansion
- Postal code normalization

Matching Strategy

Deterministic exact matching on normalized fields identifies clear duplicates. Fuzzy matching with configurable thresholds catches near-duplicates that may result from typos or intentional variation. Near-duplicate matches route to manual review rather than automatic rejection.

Device and Behavioral Signals

- Device fingerprinting to detect account farming from single devices
- Login pattern analysis for anomaly detection
- Risk scoring combining multiple signals
- Suspicious patterns trigger manual review

6.4 Verification Flow

Users initiate verification from within the SwarpPay application. The flow includes:

- Identity document capture (passport, national ID, or driver's license)

- Liveness verification to confirm the user is present
- Data extraction and validation against issuing authority databases
- Anti-duplicate screening against existing verified users
- Risk scoring and eligibility determination

Most verifications complete within minutes. Edge cases flagged by automated systems route to manual review with target resolution within 24 hours.

6.5 Privacy Considerations

Personal identification information is stored separately from general account data with enhanced access controls. Data retention follows regulatory requirements and user consent. Users can request data deletion subject to legal retention obligations. SwarpID does not share individual identity data with third parties beyond the KYC provider relationship.

6.6 Future Development

SwarpID development follows a phased roadmap:

v0 (2026 Launch): Third-Party Integration

- Third-party KYC provider for identity verification
- Third-party fiat on/off-ramp partner (MoonPay)
- Internal anti-duplicate checks and risk signal monitoring
- Manual review for edge cases

v1 (2027): SwarpID Native Identity Layer

- SwarpID replaces all third-party KYC providers
- Native fiat on/off-ramp infrastructure
- Full ownership of identity verification pipeline
- Enhanced duplicate detection and risk scoring

v2 (2028+): Advanced Features

- Proof-of-personhood style uniqueness guarantees
- Portable reputation signals across platforms
- B2B identity verification services for partners
- Cross-platform verification (subject to regulatory constraints)

7. SWARP Token Economics

7.1 Token Overview

SWARP is the native utility token of the SwarpPay ecosystem, deployed on Solana as an SPL token. The token serves multiple functions including platform fee discounts, ecosystem rewards distribution, and governance participation.

Property	Value
Token Name	SWARP
Network	Solana (SPL Token)
Decimals	6
Total Supply	1,000,000,000 SWARP
Mint Authority	Revoked after initial mint (fixed supply)
Freeze Authority	Revoked (no account freezes)

7.2 Token Distribution

Allocation	Percentage	Tokens	Vesting
Team	12%	120,000,000	12-month cliff, 36-month linear
Investors (Seed + Strategic)	15%	150,000,000	6-month cliff, 18-month linear
Advisors and Partners	3%	30,000,000	6-month cliff, 12-month linear
Ecosystem Rewards	28%	280,000,000	7-year emission schedule
Treasury and Reserve	17%	170,000,000	Multi-sig controlled, no auto release
DEX Liquidity (Protocol-Owned)	5%	50,000,000	At TGE for initial liquidity
CEX Liquidity Reserve	3%	30,000,000	Reserved for exchange listings
Community and Public Float	17%	170,000,000	50M at TGE, 120M for later programs

7.3 Vesting Schedules

Team Allocation (120,000,000 SWARP)

Team tokens are subject to a 12-month cliff from TGE. After the cliff, tokens vest linearly over 36 months. Monthly release after cliff: 120,000,000 / 36 = 3,333,333.33 SWARP.

Investor Allocation (150,000,000 SWARP)

Investor tokens have a 6-month cliff followed by 18-month linear vesting. Monthly release after cliff: 150,000,000 / 18 = 8,333,333.33 SWARP.

Advisor Allocation (30,000,000 SWARP)

Advisor tokens have a 6-month cliff followed by 12-month linear vesting. Monthly release after cliff: 30,000,000 / 12 = 2,500,000 SWARP.

7.4 Ecosystem Rewards Schedule

The 280,000,000 token ecosystem rewards allocation distributes over 7 years with decreasing annual emissions:

Year	Tokens	Percentage of Allocation
Year 1	56,000,000	20.0%
Year 2	50,400,000	18.0%
Year 3	44,800,000	16.0%
Year 4	39,200,000	14.0%
Year 5	33,600,000	12.0%
Year 6	28,000,000	10.0%
Year 7	28,000,000	10.0%

7.5 Treasury Management

The 170,000,000 token treasury reserve has no automatic release schedule. Tokens are controlled by a multi-signature wallet (target: Squads protocol on Solana) with role-based approval requirements. Treasury deployments require governance approval and monthly reporting. An annual budget cap recommendation guides responsible treasury management.

7.6 Liquidity Provisioning

Initial DEX Liquidity

At TGE, 50,000,000 SWARP will be paired with $1,500,000 equivalent in a DEX liquidity pool, implying an initial price of approximately $0.03 per SWARP and a fully diluted valuation of $30,000,000.

CEX Reserve

30,000,000 SWARP is reserved for centralized exchange listing requirements. These tokens will only be deployed as needed for specific exchange partnerships.

7.7 Buyback Policy

Starting post-break-even, SwarpPay will implement a buyback program using 10% of net protocol revenue for SWARP market purchases. Buybacks will be executed monthly with published formula, schedule, and receiving wallets for transparency.

8. Security Architecture

8.1 Authentication and Authorization

User Authentication

- Multiple authentication methods: email/password, phone/SMS OTP, social login (Google, Apple)
- All passwords hashed with bcrypt (10 salt rounds)
- JWT tokens for session management with configurable expiry
- Refresh token rotation for extended sessions

Wallet PIN

- 6-digit numeric PIN required for high-risk operations
- Hashed with bcrypt before storage
- 5 incorrect attempts trigger 30-minute lockout
- Separate from account password for defense in depth

Authorization Model

- Role-based access control for administrative functions
- Wallet ownership verification for all wallet operations
- Step-up authentication for sensitive actions

8.2 Data Protection

Encryption at Rest

- Database-level encryption via managed database service
- Deposit address private keys encrypted with AES-256-CBC
- Random initialization vectors for each encrypted value
- Encryption keys managed through environment configuration

Encryption in Transit

- TLS encryption for all API traffic
- SSL enabled for database connections
- Secure WebSocket connections where applicable

Sensitive Data Handling

- Passwords and PINs stored only as bcrypt hashes
- OTP codes stored temporarily with automatic expiry
- PII isolated with enhanced access controls

8.3 API Security

Input Validation

- Global validation pipeline using class-validator
- Type checking and constraint validation on all inputs
- Length limits, pattern matching, and enum validation

Injection Prevention

- All database queries use parameterized statements via TypeORM
- No raw SQL in application code
- QueryBuilder uses named parameters

Network Security

- DDoS protection at infrastructure layer
- CORS configuration restricting allowed origins
- Rate limiting on sensitive endpoints

8.4 Operational Security

Monitoring and Logging

- Comprehensive logging of security-relevant events
- Authentication attempts and failures
- Withdrawal requests and completions
- Risk signal triggers and manual review decisions

Incident Response

- Anomaly detection triggers automatic review/freeze
- Escalation procedures for security incidents
- Regular security review and hardening

8.5 Audit Status

SwarpPay is in active development with security hardening ongoing. Third-party security audits for both smart contracts and backend infrastructure are planned prior to mainnet launch. A bug bounty program will be established to incentivize responsible disclosure of vulnerabilities.

9. Compliance Framework

9.1 Regulatory Strategy

SwarpPay pursues a licensing-first regulatory strategy. SWARP FOUNDATION S.R.L. is incorporated in Italy; however, the Italian registration does not authorize custodial crypto trading

or fiat on/off-ramp services. Upon securing seed investment, SwarpPay will immediately file applications for two primary licenses:

Estonia: MiCA CASP Authorization

Under the Markets in Crypto-Assets (MiCA) regulation, SwarpPay will apply for Crypto-Asset Service Provider (CASP) authorization in Estonia, providing a pathway to offer services across the European Union.

Dubai: VARA VASP License

In the United Arab Emirates, SwarpPay will apply for Virtual Asset Service Provider (VASP) licensing under the Virtual Assets Regulatory Authority (VARA) framework for the Dubai market (excluding DIFC).

9.2 Interim Bridge Model

Authorization timelines are not guaranteed. Since the Italian registration does not permit custodial trading or fiat services, SwarpPay will operate through licensed third-party partners during the bridge period until CASP and VARA authorizations are obtained.

Partner Responsibilities (2026)

- KYC/AML compliance operations via licensed KYC provider
- Transaction monitoring and case handling
- Fiat on/off-ramp services via MoonPay
- Custody oversight where required by regulation

SwarpPay Responsibilities

- Product UX and technical implementation
- Trading experience and operational controls
- Customer support (front-line)
- Platform security and incident response

9.3 Compliance Controls

Transaction Monitoring

- Automated monitoring of transaction patterns
- Threshold-based flagging for review
- Sanctions list screening on withdrawals
- Risk-based manual review triggers

Record Keeping

- Comprehensive audit trail of user actions
- Transaction records with timestamps and metadata
- Immutable logging of compliance-relevant events
- Data retention aligned with regulatory requirements

Customer Due Diligence

- Tiered verification based on risk and feature access
- Third-party KYC verification for identity confirmation
- Enhanced due diligence for high-risk indicators
- Ongoing monitoring of verified users

9.4 Outsourcing Framework

SwarpPay operates under a documented outsourcing and controls framework covering:

- Clear RACI matrix for compliance, payments, incident response, and support
- Security control requirements (2FA, step-up auth, device binding, audit logs)
- Data handling standards (least privilege, encryption, retention)
- Business continuity and incident management procedures
- Customer disclosure and complaint handling workflows

9.5 Regulatory Communications

SwarpPay will not represent itself as authorized prior to authorization being granted. During the bridge period, customer-facing communications will clearly identify which services are provided by licensed partners and which are provided by SwarpPay.

10. Technical Infrastructure

10.1 Backend Stack

Component	Technology
Runtime	Node.js (LTS)
Framework	NestJS
Language	TypeScript
Database	PostgreSQL
ORM	TypeORM
Authentication	JWT + bcrypt
Validation	class-validator

10.2 Database Architecture

Core Entities

- Users: Authentication credentials, profile data, verification status
- Wallets: Public keys, encrypted private keys, balances, status
- Transactions: Type, status, amounts, blockchain references
- Rewards: Type, section, progress, claim status
- Launchpad Projects: Token details, status, raise progress, graduation data

Indexing Strategy

- Unique indexes on email, phone, username, wallet address
- Foreign key indexes for join performance
- Status indexes for filtered queries
- Composite indexes for common query patterns

10.3 Blockchain Integration

Solana Connection

- Configurable RPC endpoint (supports multiple providers)
- Connection pooling for efficient resource use
- Confirmation level: finalized for accounting, confirmed for UI updates

DEX Integration

- Jupiter aggregator for optimal swap routing
- Raydium SDK for direct pool access
- Quote fetching with slippage calculation
- Transaction construction with compute budget optimization

Transaction Monitoring

- Polling-based signature monitoring (30-second intervals)
- Confirmation tracking to finalized status
- Automatic retry for transient failures
- Webhook notifications on status changes

10.4 External Service Integration

Service Category	Purpose	Provider (2026)
Identity Verification	User KYC and document verification	Third-party (SwarpID in 2027)
Fiat On/Off-Ramp	Fiat currency to crypto conversion	MoonPay (SwarpID in 2027)
Email Service	Transactional emails and notifications	SendGrid
SMS Service	OTP delivery and alerts	Twilio
Push Notifications	Mobile app notifications	Firebase

Price Oracles	Real-time crypto price data	DexScreener, CoinGecko
File Storage	Profile pictures and documents	AWS S3

10.5 Deployment Architecture

SwarpPay is deployed on cloud infrastructure with separated services for API, database, and sensitive operations. The architecture supports horizontal scaling to accommodate user growth.

Environment Separation

- Development: Local development with devnet Solana
- Staging: Pre-production testing environment
- Production: Live environment with mainnet-beta Solana

Database Configuration

- Managed PostgreSQL with automated backups
- Point-in-time recovery capability
- SSL connections enforced

Secrets Management

- Environment-based configuration
- Encryption keys stored securely outside codebase
- Access controls on sensitive configuration

11. Appendix

11.1 Glossary

Term	Definition
Bonding Curve	Mathematical function determining token price based on supply sold
CASP	Crypto-Asset Service Provider under MiCA regulation
Custodial	Service model where platform manages private keys on behalf of users
DAMM	Dynamic Automated Market Maker (Meteora protocol)
DEX	Decentralized Exchange
FDV	Fully Diluted Valuation
Graduation	Transition from bonding phase to open trading on DEX
HSM	Hardware Security Module
KYC	Know Your Customer identity verification
LP	Liquidity Provider or Liquidity Pool
MiCA	Markets in Crypto-Assets (EU regulation)
MPC	Multi-Party Computation
OTP	One-Time Password
PDA	Program Derived Address (Solana)
RPC	Remote Procedure Call (blockchain node interface)
SPL	Solana Program Library (token standard)
Sybil Attack	Attack using multiple fake identities
TGE	Token Generation Event
VARA	Virtual Assets Regulatory Authority (Dubai)
VASP	Virtual Asset Service Provider

11.2 Contact Information

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11.3 Document Information

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End of Document